

Chains of reasoning - Set 1 - Solutions

Specification Prf2, Prf3, Prf5.

Answer key

Question	Answer	Topic
1	A	Chains of reasoning
2	B	Chains of reasoning
3	A	Chains of reasoning
4	D	Chains of reasoning
5	F	Chains of reasoning
6	E	Chains of reasoning
7	D	Chains of reasoning
8	D	Chains of reasoning
9	D	Chains of reasoning
10	E	Chains of reasoning
11	C	Chains of reasoning
12	F	Chains of reasoning

Worked solutions

Question 1 RSN-05 · Chains of reasoning · spec Prf5

In a knockout tournament with 37 players, each match eliminates exactly one player and the tournament ends when one player remains. How many matches are played?

Answer: A

Each match eliminates one player and 36 must be eliminated, so 36 matches - whatever the draw looks like.

Full method - how and why

An invariant argument: count what each match achieves rather than simulating the draw.

- 1 At the end one player remains, so $37 - 1 = 36$ players have been eliminated.

why: Count the outcome, not the structure.

- 2 Each match eliminates exactly **one** player.

why: Given in the question - this is the invariant.

- 3 So the number of matches equals the number of eliminations: 36.

why: And this holds whatever byes or bracket shape are used, which is why the answer is unique.

Where it goes wrong

- Trying to build the bracket and getting lost in byes because 37 is not a power of 2.
- Halving repeatedly (18, 19-type answers), which only works for tidy brackets.

Takeaway Count the invariant: a knockout with n players always takes $n - 1$ matches.

Question 2 RSN-07 · Chains of reasoning · spec Prf5

The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?

Answer: B

The three rows together contain every digit once, so $3 \times (\text{row sum}) = 1 + 2 + \dots + 9 = 45$, giving 15.

Full method - how and why

Count the same total two different ways - the standard double-counting move.

- 1 The three rows between them contain each digit 1 to 9 exactly once.

why: Every cell lies in exactly one row, so the rows partition the grid.

- 2 So the three row sums add to $1 + 2 + \dots + 9 = 45$.

why: Use $\frac{9 \times 10}{2} = 45$.

- 3 All three row sums are equal, so each is $\frac{45}{3} = 15$.

why: And the columns give the same total, so they must each be 15 too - consistent.

Where it goes wrong

- Answering 45, the grand total rather than one row's sum.
- Trying to construct an actual magic square; the total settles it with no construction needed.

Takeaway Double-count a total: rows partition the grid, so $(\text{number of rows}) \times (\text{row sum}) = \text{grand total}$.

Question 3 RSN-10 · Chains of reasoning · spec Prf5

Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?

Answer: A

The 50 pairs $\{1, 100\}, \{2, 99\}, \dots, \{50, 51\}$ each sum to 101. Fifty cards (one from each pair) may avoid it; 51 cannot.

Full method - how and why

Pigeonhole again, with the pairs built to sum to the target.

- 1 Pair the numbers to sum to 101: $\{1, 100\}, \{2, 99\}, \dots, \{50, 51\}$.

why: Each of 1 to 100 lies in exactly one such pair, so there are 50 pairs.

- 2 Drawing one card from each pair gives 50 cards with **no** pair summing to 101 - for example 1 to 50.

why: So 50 is not enough; this constructs the worst case.

- 3 Any 51 cards must include two from one pair, and those sum to 101.

why: 51 cards into 50 pairs forces a repeat.

Where it goes wrong

- Answering 50 - the largest number that can fail.
- Answering 2 by thinking about luck rather than certainty.

Takeaway Same recipe as any pigeonhole guarantee: build the pairs, show the worst case, then add one.

Question 4 RSN-14 · Chains of reasoning · spec Prf5

In a class every student studies at least one of Maths and Physics. 70% study Maths and 60% study Physics. A student is picked at random. What is the probability they study both?

Answer: D

$P(M \text{ or } P) = 1 = 0.7 + 0.6 - P(\text{both})$, so $P(\text{both}) = 0.3$.

Full method - how and why

The addition rule, with 'every student studies at least one' supplying the missing total.

- 1 'At least one of Maths and Physics' means $P(M \text{ or } P) = 1$.

why: Nobody is outside the union, so the union has probability 1.

- 2 Apply the addition rule: $P(M \cup P) = P(M) + P(P) - P(M \cap P)$.

why: Subtract the overlap, which the two percentages each count.

- 3 Substitute: $1 = 0.7 + 0.6 - P(\text{both})$, so $P(\text{both}) = 1.3 - 1 = 0.3$.

why: The excess of 1.3 over 1 is exactly the double-counted overlap.

Where it goes wrong

- Multiplying $0.7 \times 0.6 = 0.42$ by assuming independence - nothing here says the subjects are independent, and the union condition rules it out.
- Answering 'cannot be determined'; the 'at least one' condition is precisely what makes it determinable.

Takeaway When percentages sum to more than 100% and everyone is covered, the excess is the overlap.

Question 5 RSN-16 · Chains of reasoning · spec Prf5

The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?

Answer: F

$(m-4) + (m-2) + m + (m+2) + (m+4) = 5m$ where m is the middle (odd) number; $5m$ is odd, so 10 is never a factor.

Full method - how and why

Represent the numbers symmetrically about the middle so everything cancels, then read off the guaranteed factor.

- 1 Let the middle number be m (odd). The five are $m-4, m-2, m, m+2, m+4$.

why: Consecutive odd numbers differ by 2; centring makes the offsets cancel in pairs.

- 2 Add: the ± 2 and ± 4 cancel, leaving $S = 5m$.

why: So 5 is always a factor.

- 3 Is 10 always a factor? m is odd, so $5m$ is odd and never divisible by 2. **No.**

why: For instance $1 + 3 + 5 + 7 + 9 = 25$.

- 4 Are 3, 15 or 25 always factors? 25 needs $5 \mid m$, 3 needs $3 \mid m$ - neither is guaranteed ($m = 7$ gives $S = 35$).

why: Only the factor 5 survives every case.

Where it goes wrong

- Answering 25 from the single example $1 + 3 + 5 + 7 + 9 = 25$ - that is $5m$ with $m = 5$, a coincidence.
- Answering 10 by assuming an even total; five odd numbers sum to an odd number.

Takeaway An odd count of equally spaced terms sums to $(\text{count}) \times (\text{middle term})$. Test the extra factors against a second example.

Question 6 RSN-21 · Chains of reasoning · spec Prf5

Seven consecutive integers are written down. Which of the following statements is/are true?

- I Their sum is divisible by 7.
- II Their product is divisible by 7.
- III Their sum is odd.

Answer: E

The sum is $7m$ where m is the middle integer (I). Seven consecutive integers cover every remainder mod 7, so one is a multiple of 7 (II). $7m$ is odd only when m is odd, so III need not hold.

Full method - how and why

Seven consecutive integers: the sum telescopes about the middle, and the seven values cover all remainders mod 7.

- 1 Let the middle be m ; the seven are $m-3, \dots, m+3$, and the offsets cancel, so the sum is $7m$. **I is true.**

why: Symmetric representation makes the cancellation immediate.

- 2 II: seven consecutive integers take all seven remainders on division by 7, so exactly one is a multiple of 7; hence 7 divides the product. **True.**

why: Among any k consecutive integers there is exactly one multiple of k .

- 3 III: the sum is $7m$, which is odd only when m is odd. Take 1, ..., 7: sum 28, even. **Not always true.**

why: One counterexample settles it.

- 4 So I and II.

why: Match to the list.

Where it goes wrong

- Accepting III after testing 2, ..., 8 (sum 35, odd) and no other case.
- Confusing the sum and product claims - I is about $7m$, II about a multiple of 7 among the factors.

Takeaway k consecutive integers: their sum is $k \times \text{middle}$ (for odd k), and exactly one of them is a multiple of k .

Question 7 RSN-22 · Chains of reasoning · spec Prf5

A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?

Answer: D

Each of the 4 rows contains exactly 2 black squares: 8 in total.

Full method - how and why

Count by rows - the column condition is extra information, not needed for the count.

- 1 There are 4 rows and each contains exactly 2 black squares.
why: Every black square lies in exactly one row, so rows partition them.
- 2 Total $= 4 \times 2 = 8$.
why: No double counting.
- 3 Check with the columns: $4 \times 2 = 8$ as well - consistent, which is why such a colouring can exist.
why: If the two counts disagreed, no such grid would exist.

Where it goes wrong

- Adding rows and columns to get 16 - that counts every black square twice.
- Answering 4 by counting rows rather than squares.

Takeaway Count via one partition only; use the other as a consistency check, not as extra squares.

Question 8 RSN-01 · Chains of reasoning · spec Prf5

A selection of n distinct integers is taken from $1, 2, \dots, 20$. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

Answer: D

Pair up $\{1, 11\}, \{2, 12\}, \dots, \{10, 20\}$: ten pairs. Choosing 11 numbers forces two from the same pair (pigeonhole); choosing one from each pair gives 10 numbers with no such difference.

Full method - how and why

A pigeonhole question. Build the 'boxes' so that two items in one box give exactly what you want, then answer boxes + 1.

- 1 Pair the numbers so that each pair differs by 10: $\{1, 11\}, \{2, 12\}, \dots, \{10, 20\}$.
why: Every number from 1 to 20 lies in exactly one pair, and two numbers differ by 10 precisely when they share a pair.
- 2 There are 10 pairs, so 10 numbers can be chosen with no two in the same pair - for example 1 to 10.
why: This shows 10 is **not** enough, which is the half of the argument people skip.
- 3 With 11 numbers and only 10 pairs, two must come from the same pair - and those differ by 10.
why: Pigeonhole: more items than boxes forces a repeat.
- 4 So the smallest guaranteed n is 11.
why: Boxes + 1.

Where it goes wrong

- Answering 10 - the largest selection that can **avoid** the condition, not the smallest that forces it.
- Answering 2, confusing 'could happen' with 'guaranteed'.
- Forgetting to show 10 is achievable; without it you have not proved 11 is smallest.

Takeaway 'Smallest n that guarantees' = (largest bad selection) + 1. Construct the boxes so a shared box means success.

Question 9 RSN-03 · Chains of reasoning · spec Prf5

Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?

Answer: D

If four people each shook hands with all four others, everyone shook hands with everyone, so the fifth person also made 4 handshakes, not 1. (The total $4 + 4 + 4 + 4 + 1 = 17$ is odd, which is a second reason.) The others can happen: 'exactly two each' is a pentagon of handshakes, 'exactly four each' is everyone with everyone, 1, 1, 2, 2, 2 is a chain, and one person shaking no hands just leaves them out.

Full method - how and why

A handshake (graph) consistency question. Test each list for an internal contradiction.

- 1 Take 4, 4, 4, 4, 1: four people each shook hands with all four others.
why: With five people, the maximum possible is 4 handshakes each.
- 2 If person A shook hands with everyone, that includes the fifth person. The same for the other three.
why: Each of the four 'maximum' people contributes one handshake to the fifth person.
- 3 So the fifth person made at least 4 handshakes, not 1. **Impossible.**
why: The list contradicts itself.
- 4 The others are all realisable: 2, 2, 2, 2, 2 is a pentagon; 4, 4, 4, 4, 4 is everyone with everyone; 1, 1, 2, 2, 2 is a chain $A-B-C-D-E$; and 0 for one person just leaves them out.
why: Constructing one arrangement is enough to show a list is possible.

Where it goes wrong

- Rejecting 'everyone shakes four hands' as too many - it is exactly the complete case and is fine.
- Rejecting the list 1, 1, 2, 2, 2 without trying to build it; a chain works.
- Forgetting the parity check: the degree total must be even. $4 + 4 + 4 + 4 + 1 = 17$ is odd, which is a second, instant reason it is impossible.

Takeaway Two fast checks on a handshake list: the total must be even, and someone shaking everyone's hand forces everyone else's count up.

Question 10 RSN-04 · Chains of reasoning · spec Prf5

x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

Answer: E

$(x - y)(x + y) = 45$ with $0 < x - y < x + y$: (1, 45), (3, 15), (5, 9) give $(x, y) = (23, 22), (9, 6), (7, 2)$. Three pairs.

Full method - how and why

A difference of two squares turns a quadratic condition into a factor-pair search.

- 1 Factorise: $x^2 - y^2 = (x - y)(x + y) = 45$.
why: Difference of two squares converts the equation into a product of two integers.
- 2 Both factors are positive with $x - y < x + y$, so list factor pairs of 45 with the smaller first: (1, 45), (3, 15), (5, 9).
why: x, y positive forces $x + y > 0$, and then $x - y > 0$ too since the product is positive.
- 3 Solve each: $x = \frac{\text{sum}}{2}$, $y = \frac{\text{difference}}{2}$, giving (23, 22), (9, 6), (7, 2).
why: Adding the two equations gives $2x$; subtracting gives $2y$. Both factors are odd, so both results are integers.
- 4 All three have positive integer y , so there are **three** pairs.
why: Check one: $9^2 - 6^2 = 81 - 36 = 45$ ✓.

Where it goes wrong

- Including (9, 5) or other reversed pairs - $x - y$ must be the smaller factor.
- Forgetting to check that x and y come out as integers; if 45 had a factor pair of mixed parity it would have to be discarded.
- Counting $45 = 45 \times 1$ and 1×45 as two different solutions.

Takeaway $x^2 - y^2 = N$: list factor pairs $N = de$ with $d < e$ and the same parity, then $x = \frac{d+e}{2}$, $y = \frac{e-d}{2}$.

Question 11 RSN-06 · Chains of reasoning · spec Prf5

Three statements are made about a positive integer N :

- N is a perfect square.
- N is a multiple of 6.
- N is less than 50.

Exactly two of the statements are true. Which of the following could NOT be N ?

Answer: C

36 makes all three statements true, not exactly two. Each of the others makes exactly two true: 16 and 25 (square, < 50), 48 and 12 (multiple of 6, < 50), 144 (square, multiple of 6).

Full method - how and why

Test each candidate against all three statements and count how many are true. You want the one that does **not** give exactly two.

- 1 36: perfect square (6^2) ✓, multiple of 6 ✓, less than 50 ✓ - **three** true.

why: The condition demands exactly two, so 36 is ruled out.

- 2 16: square ✓, multiple of 6 ✗, < 50 ✓ - two. Allowed.

why: Check each of the three statements, do not stop early.

- 3 25: square ✓, multiple of 6 ✗, < 50 ✓ - two. 48: square ✗, multiple of 6 ✓, < 50 ✓ - two. 12: the same pattern - two.

why: All fine.

- 4 144: square (12^2) ✓, multiple of 6 ✓, < 50 ✗ - two. Allowed.

why: So only 36 fails.

Where it goes wrong

- Reading 'exactly two are true' as 'at least two', which would admit 36.
- Missing that $144 = 12^2$ is both a square and a multiple of 6.
- Looking for the value making the fewest statements true instead of the most.

Takeaway 'Exactly k ' excludes $k + 1$ as firmly as $k - 1$; tabulate every statement for every candidate.

Question 12 RSN-08 · Chains of reasoning · spec Prf5

a , b and c are real numbers with $a < b < c$. Which of the following statements must be true?

I $a + b < 2c$

II $ab < bc$

III $\frac{a+c}{2} < c$

Answer: F

I: $a + b < c + c$. III: $\frac{a+c}{2} < \frac{c+c}{2} = c$. II fails when b is negative: $a = -3$, $b = -2$, $c = -1$ gives $ab = 6 > bc = 2$.

Full method - how and why

Inequality manipulation with 'must be true'. Two follow from replacing a term by a larger one; the third involves multiplying, which needs sign information.

1 I: from $a < c$ and $b < c$, add: $a + b < 2c$. **True.**

why: Adding two inequalities of the same direction is always valid.

2 III: $a < c$ so $a + c < 2c$, and dividing by 2 gives $\frac{a+c}{2} < c$. **True.**

why: Dividing by a positive constant preserves the direction. (The midpoint of two numbers lies below the larger.)

3 II: $ab < bc$ would follow from $a < c$ only if we could multiply by b - valid only for $b > 0$.

why: Multiplying an inequality by a negative number reverses it.

4 Counterexample: $a = -3$, $b = -2$, $c = -1$ satisfies $a < b < c$, but $ab = 6$ and $bc = 2$, so $ab > bc$. **False.**

why: All-negative values are the natural place to test a multiplication claim.

5 So I and III.

why: Match to the list.

Where it goes wrong

- Accepting II by 'multiplying $a < c$ by b ' without checking the sign of b .
- Testing only positive numbers, where II does hold.

Takeaway Adding inequalities is always safe; multiplying by a variable is not - you need its sign.